Chapter 2

STEADY STATE CONDUCTION IN ONE DIMENSION

Conduction of heat means transport of heat energy in a medium from a region at a higher temperature to a region at a lower temperature without any macroscopic motion in the medium. The difference in temperature between the regions causes the flow of heat and is called the *temperature driving force.* Heat conduction is also called *diffusion of heat.*

The mechanism of heat conduction in a medium depends upon the state of the medium, i.e. whether it is a solid, a liquid or a gas. In a solid, the molecular motion is restricted to vibrations about an equilibrium position. In the presence of a temperature gradient heat energy is transferred from one molecule to a neighbouring molecule through molecular vibrations. In metals, however, conduction of heat occurs more through the drift of free electrons than by molecular vibrations. The motion of free electrons in metals is similar to that of molecules in a gas (free electrons are often referred to as *electron gas),* and this is why a material having good electrical conductivity also possesses good thermal conductivity.

In a gaseous medium, conduction of heat occurs through collisions of molecules having more thermal energy (i.e. faster moving) with molecules having less thermal or kinetic energy (slower moving). While similar phenomenon is partly responsible for heat conduction in a liquid, there are other factors too which involve intermolecular forces in the liquid.

Transport of heat in a solid occurs only by conduction. In this chapter we will describe the basic law of heat conduction and discuss its applications to heat transfer calculations in single or multi-layer solids of three common geometries - plane wall, cylindrical and spherical.

2.1. THE BASIC LAW OF HEAT CONDUCTION - FOURIER'S LAW

The basic law of heat conduction in a medium was established by J.B.J. Fourier, a physicist, in 1822 from his experimental data on the rate of heat flow. The law states that if *two plane parallel surfaces each having an area A are separated by a distance l and are maintained at temperatures T_1 and T_2, respectively ($T_1 > T_2$)* (Fig. 2.1), *the rate of heat conduction Q at steady state through the wall is given by*

$$Q = kA\frac{T_1 - T_2}{l} \tag{2.1}$$

where k is called the thermal conductivity of the solid and is assumed to be constant throughout the wall.

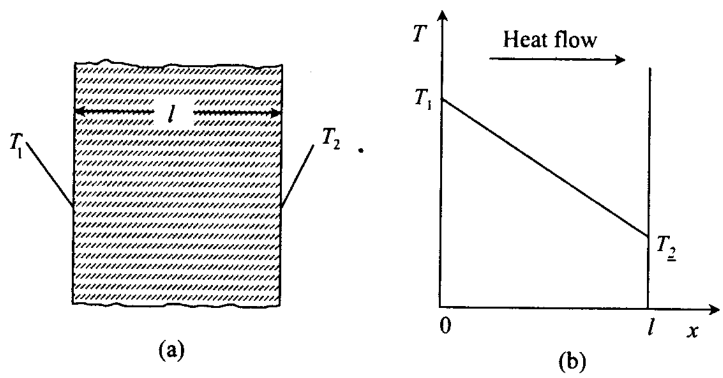


Fig. 2.1: Steady state conduction of heat in a plane wall: (a) the wall and (b)the temperature profile

In the differential form the Fourier's law is expressed as

$$q_x = -k\frac{dT}{dx} \qquad (2.2)$$

where q_x is the heat flux (i.e. the rate of heat conduction in the x-direction per unit area normal to the x-direction**),** and dT/dx *is* the temperature gradient in the x-direction. Heat flow in the direction of decreasing temperature is a positive quantity, therefore dT/dx *is* negative. The negative sign in Eq. (2.2) makes both sides of it consistent with respect to sign. The units of the various quantities in Eqs. (2.1) and (2.2) are: Q, W (watt) [or, kcal/s, Btu/h, etc.] ; k, W/m °C [or, cal/s cm °C, Btu/h ft °F, etc.] ; q_x, W/m^2 [or, kcal/s cm^2, Btu/h ft^2, etc.] ; dT/dx, °C/m [or, K/m, °F/ft, etc.]. The quantities q_x and dT/dx are vector quantities. Equation (2.1) is the integrated form of Eq. (2.2) over the thickness l of the wall for an area A.

2.2. THERMAL CONDUCTIVITY

Thermal conductivity is a fundamental property of a material that gives a measure of the effectivity of the material in transmitting heat through it. Besides its chemical constitution, characteristics of a material (solid, liquid or gas), nature of the solid state (crystalline or amorphous), and physical conditions (temperature, pressure) have significant effects on thermal conductivity.

A material in the crystalline state has a higher thermal conductivity than that of the same material in the amorphous state. The orderly arrangement of the atoms in a crystalline solid allows faster transmission of thermal energy through vibrations of the crystal lattices. The thermal conductivity of a substance depends to some extent on temperature. For metals, thermal conductivity generally decreases with an increase in temperature (aluminium is an exception). A metal in the pure state has the maximum thermal conductivity. The thermal conductivity decreases with increasing amounts of impurities in a metal. Most nonmetallic substances are poor conductors of heat.

Solid substances having low thermal conductivities are called ***thermal insulators***. Such a substance can be applied to the surface of a metal to reduce its rate of exchange of heat energy with the surroundings. Besides low thermal conductivity, an insulator must be chemically inert, available in a form suitable for application on a surface, and should be light and cheap. Commercial insulators are ceramic (e.g. insulating bricks), diatomaceous (e.g. rock wool) or polymeric (e.g. expanded polyurethane, expanded polystyrene, etc.) materials. These materials are either highly porous, or have a high void volume filled with an inert gas. A gas being a very poor conductor of heat, a high void volume fraction makes an insulator more effective in reducing the flow of heat. In fact, the presence of entrapped gas is more responsible for the insulating property than for the ‘true’ thermal conductivity of the material. The thermal conductivity of an insulating material increases with temperature.

It is relevant here to point out the difference between an insulating material and a refractory. The latter is a substance capable of withstanding high temperatures without physical deterioration, but need not necessarily have a low thermal conductivity like an insulator. However in most cases, low thermal conductivity is a desirable characteristic of refractory materials. A refractory material must have chemical resistance to the environment to which it is exposed. A furnace, for example, is provided with an innermost layer of refractory bricks, one or two layers of insulating bricks, and often with an outer layer of ordinary bricks. Common examples of refractory materials are silica, magnesite, alumina, fire-clay, sillimanite (Al_2Si0_5), chromite ($FeO.Cr_20_3$), zirconia (ZrO_Z), etc. For most refractories, the thermal conductivity increases with increase in temperature (exception: magnesite bricks).

Thermal conductivities of liquids generally decrease with increase in temperature (exception: glycerine, water, etc. over certain ranges of temperature), but are nearly independent of pressure. Leaving aside liquid metals, water has the highest thermal conductivity of all liquids. Thermal conductivities of gases increase with temperature and decrease with molecular weight; pressure dependence becomes significant only at high pressures. Among the gases, hydrogen has the highest thermal conductivity.

Generally, gases have lower thermal conductivities than those of liquids, and liquids have conductivties lower than those of solids. For example, at the ambient temperature air has a thermal conductivity of 0.0262 W/m K (0.0151 Btu/h ft °F), water has a thermal conductivity of 0.63 W/m K (0.364 Btu/h ft °F), whereas pure silver has a conductivity of 410 W/m K (237 Btu/h ft °F).

In heat transfer calculations, it is often satisfactory and sufficient to take the thermal conductivity value of a substance at the average temperature of the material or medium. However, a linear or a quadratic equation as given below can be used to describe thermal conductivity as a function of temperature.

$$k = k_o(1 + aT + bT^2) \tag{2.3}$$

where a and b are constant coefficients, and k_o is the thermal conductivity at T = 0 K.

If the thermal conductivity of a solid is the same in all directions, the material is called isotropic. But there are some materials in which the conductivity depends upon the direction as well. Such materials are called *anisotropic.* For example, the thermal conductivity of wood in the direction of the grains is different from that in the transverse direction. So wood is an anisotropic material.

Thermal conductivities of some substances are given below in Table 2.1.

Table 2.1 Thermophysical properties of some common materials

Material	*Temperature* (°C)	*Density* (kg/m³)	*Specific heat* (kJ/kg °C)	*Thermal conductivity* (W/m °C)
Non-metals				
Building bricks	20	1600	0.84	0.69
Fireclay brick (fired at 1450°C)	500	2300	0.96	1.28
Magnesite	1200	—	—	1.90
Window glass	20	2700	0.84	0.78
Asbestos sheet	59	—	—	0.166
Corkboard (density = 160 kg/m³)	30	—	—	0.043
Wool	30	130–200	—	0.036
Metals				
Carbon steel (0.5% C)	100	7800	0.465	52
Stainless steel (316)	27	8240	0.468	13.4
Stainless steel (304)	27	7900	0.477	14.9
Copper (pure)	27	8933	0.385	401
Aluminium (pure)	27	2702	0.903	237
Silver (99.9%)	20	10,525	0.234	407
Zinc (pure)	20	7144	0.384	112.2

2.3. STEADY STATE CONDUCTION OF HEAT THROUGH A COMPOSITE SOLID

For a 'single-layered' plane wall, the rate of heat conduction Q can be calculated directly from Eq. (2.1), if the surface area A, the wall thickness l, the thermal conductivity k, and the temperature difference are known. We can extend the same equation to the case of a composite or 'multi-layered' wall.

Let us consider a composite wall consisting of three layers (Fig. 2.2) of materials 1, 2 and 3, having thicknesses l_1, l_2 and l_3, and thermal conductivities k_1, k_2 and k_3, respectively. The . boundary temperatures of the different layers are shown in Fig. 2.2.

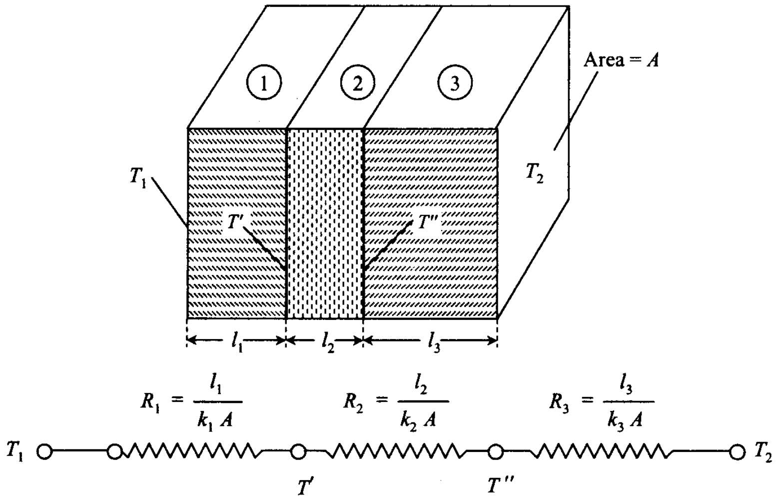


Fig. 2.2 Heat conduction in a composite wall and its electrical analogue.

The area of heat conduction A is constant. Therefore, the rates of heat flow at steady state through the individual layers are equal ($Q_1 = Q_2 = Q_3 = Q$, say). If the thermal conductivities of the layers are independent of temperature, the temperature distribution in each layer must be linear (but of different slopes). The rates of heat flow through the walls as given by Fourier's law are as follows:

Layer 1: $$Q = \frac{k_1 A(T_1 - T')}{l_1}; \qquad T_1 - T' = Q\frac{l_1}{k_1 A} \tag{2.4}$$

Layer 2: $$Q = \frac{k_2 A(T' - T'')}{l_2}; \qquad T' - T'' = Q\frac{l_2}{k_2 A} \tag{2.5}$$

Layer 3: $$Q = \frac{k_3 A(T'' - T_2)}{l_3}; \qquad T'' - T_2 = Q\frac{l_3}{k_3 A} \tag{2.6}$$

Adding Eqs. (2.4), (2.5) and (2.6), we have

$$T_1 - T_2 = Q\left(\frac{l_1}{k_1 A} + \frac{l_2}{k_2 A} + \frac{l_3}{k_3 A}\right)$$

or

$$Q = \frac{T_1 - T_2}{\frac{l_1}{k_1 A} + \frac{l_2}{k_2 A} + \frac{l_3}{k_3 A}} \tag{2.7}$$

In Eq. (2.7), $(T_1 - T_2)$ is the overall temperature driving force that causes a rate of heat transfer Q. This equation, in general, means

$$\text{Rate of heat transfer} = \frac{\text{Temperature driving force}}{\text{Thermal resistance}} \tag{2.8}$$

The analogy of the above relation with the flow of current through an electrical conductor is apparent.

$$\text{Current} = \frac{\text{Potential difference}}{\text{Electrical resistance}}$$

From Eqs. (2.4)-(2.7) and Eq. (2.8), we may write

$$\text{Thermal resistance of layer 1:} \quad R_1 = \frac{l_1}{k_1 A} \tag{2.9a}$$

$$\text{Thermal resistance of layer 2:} \quad R_2 = \frac{l_2}{k_2 A} \tag{2.9b}$$

$$\text{Thermal resistance of layer 3:} \quad R_3 = \frac{l_3}{k_3 A} \tag{2.9c}$$

The thermal resistance R_T of the composite wall is given by

$$R_T = R_1 + R_2 + R_3 = \frac{l_1}{k_1 A} + \frac{l_2}{k_2 A} + \frac{l_3}{k_3 A} = \frac{T_1 - T_2}{Q}$$

It means that thermal resistances in series are additive as in the case of electrical resistances in series. The electrical analogue of heat conduction through the wall is also shown in Fig. 2.2.

2.4. STEADY STATE HEAT CONDUCTION THROUGH A VARIABLE AREA

In the case of a plane wall the area for heat flow is constant. However, there are solids of other geometries in which the area for heat flow is variable. Two common geometries of practical importance are cylindrical and spherical, in both of which the area depends upon the radius or the radial position. Here, we develop equations for heat flow in cylindrical and spherical bodies.

2.4.1. The Cylinder

Let us consider a hollow cylinder of inside radius r_i , outside radius r_o and length L (Fig. 2.3). The inner and outer curved surfaces are maintained at temperatures T_i and T_o, respectively. Heat flow occurs in the radial direction. The area of heat flow varies from $2\pi\, r_i L$ (inside) to $2\pi\, r_o L$ (outside).

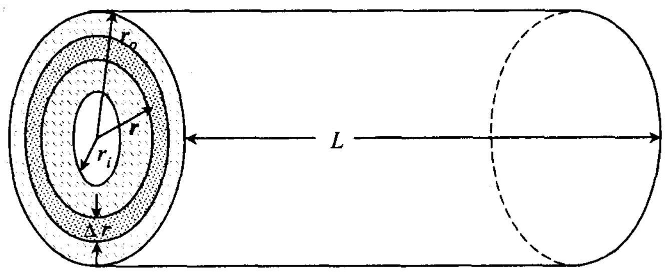


Fig. 2.3 Radial heat conduction through a thin cylindrical shell

We can make a 'heat balance' over a thin cylindrical shell of inside radius r and thickness Δr as shown in Fig. 2.3.

Rate of heat input to the thin shell at the radial position (r) = (Area)(Flux) = ($2\pi\, r$L) $q_r|_r$

Here q_r denotes the radial heat flux, and the notation '$|_r$' means that the quantity is evaluated at the radial position r.

Similarly,

Rate of heat output from the shell at the radial position r + Δr = ($2\pi\, r$L) $q_r|_{r+\Delta r}$

At steady state, there cannot be any accumulation of heat in the element, and so the rate of heat input must be equal to the rate of heat output. Therefore,

$$(2\pi rL)\, q_r|_{r+\Delta r} = (2\pi rL)\, q_r|_r$$

or

$$\frac{(2\pi rL) q_r|_{r+\Delta r} - (2\pi rL) q_r|_r}{\Delta r} = 0$$

Taking the limit $\Delta r \to 0$ and invoking the definition of the derivative of a function, we may write,

$$\frac{d}{dr}(2\pi rLq_r) = 0$$

or

$$\frac{d}{dr}(rq_r) = 0 \qquad (2.10)$$

Integrating, we get

$$r\, q_r = \text{constant} = C_1 \text{ (say)} \qquad (2.11)$$

The radial heat flux is expressed as

$$q_r = -k\frac{dT}{dr}$$

Then,

$$r\left(-k\frac{dT}{dr}\right) = C_1$$

or

$$\frac{dT}{dr} = -\frac{C_1}{k}\frac{1}{r}$$

Integrating again, we have

$$T = -\frac{C_1}{k}\ln r + C_2 \qquad (2.12)$$

here C_2 is an integration constant. The values of C_1 and C_2 can be determined by using the 'boundary conditions', that is, the known temperatures at the boundaries r_i and r_o of the solid.

$$T = T_i \quad \text{at} \quad r = r_i, \qquad T = T_o \quad \text{at} \quad r = r_o$$

Substituting in Eq. (2.12), we get

$$T_i = -\frac{C_1}{k}\ln r_i + C_2 \quad \text{and} \quad T_o = -\frac{C_1}{k}\ln r_o + C_2$$

The above two equations can be solved for C_1 and C_2.

$$C_1 = \frac{(T_i - T_o)k}{\ln(r_o/r_i)} \quad \text{and} \quad C_2 = T_i + \frac{(T_i - T_o)\ln r_i}{\ln(r_o/r_i)}$$

Substituting for C_1 and C_2 in Eq. (2.12), we get

$$T = T_i - \frac{(T_i - T_o)}{\ln(r_o/r_i)} \ln(r/r_i) \tag{2.13}$$

Equation (2.13) gives the temperature distribution in the cylinder. The rate of heat flow can be easily obtained from Eq. (2.11) which essentially means that the product of area and radial heat flux is a constant.

Rate of heat flow is given by

$$Q = 2\pi r L q_r = 2\pi L C_1 = \frac{2\pi L k(T_i - T_o)}{\ln(r_o/r_i)} \tag{2.14}$$

Comparing Eq. (2.14) with Eq. (2.8), we get

$$\text{Driving force} = (T_i - T_o); \qquad \text{Thermal resistance} = \frac{\ln(r_o/r_i)}{2\pi L k}$$

We may thus rewrite Eq. (2.14) in the form

$$Q = k\, 2\pi \frac{(r_o - r_i)}{\ln(r_o/r_i)}\, L\, \frac{(T_i - T_o)}{(r_o - r_i)} \tag{2.15}$$

The distance through which conduction occurs is the thickness of the wall of the cylinder, i.e. $r_o - r_i$. If we compare Eq. (2.15) with Eq. (2.1), the rate of heat conduction is found to be the same as that through a plane wall of thickness $(r_o - r_i)$ and area $2\pi\,[(r_o - r_i)/\ln(r_o/r_i)]\,L$. This area is called the 'log mean area of the cylinder' because it is calculated on the basis of the log mean cylinder radius, r_M. Therefore,

$$r_M = \frac{r_o - r_i}{\ln(r_o/r_i)}$$

Now we consider heat conduction through a composite cylindrical wall consisting of three layers denoted by 1, 2 and 3 having thermal conductivities k_1, k_2 and k_3 and having inner radii r_i, r' and r'', respectively. The outer radius of the composite cylinder is r_o. The temperatures at these radial positions are T_i, T', T'' and T_o, respectively. A cross-section of the assembly is shown in Fig. 2.4.

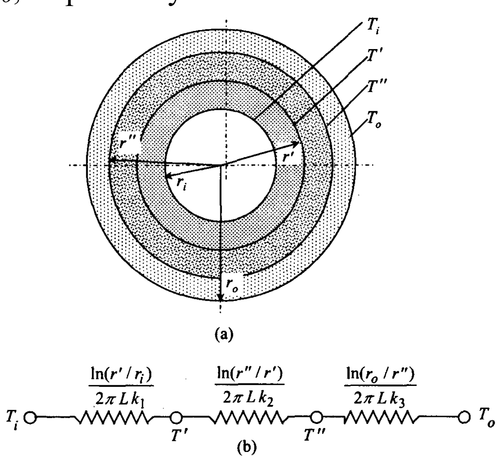


Fig. 2.4 (a) Cross-section of the composite cylinder and (b) the electrical analogue of the heat transfer resistances.

The rates of heat flow through the individual layers (which will be equal at steady state) are given by

$$Q = \frac{2\pi Lk_1(T_i - T')}{\ln(r'/r_i)} = \frac{2\pi Lk_2(T' - T'')}{\ln(r''/r')} = \frac{2\pi Lk_3(T'' - T_o)}{\ln(r_o/r'')}$$

or

$$T_i - T' = \frac{Q\ln(r'/r_i)}{2\pi Lk_1}; \quad T' - T'' = \frac{Q\ln(r''/r')}{2\pi Lk_2}; \quad T'' - T_o = \frac{Q\ln(r_o/r'')}{2\pi Lk_3} \tag{2.16}$$

Adding the above equations and rearranging, we get

$$Q = \frac{T_i - T_o}{\dfrac{\ln(r'/r_i)}{2\pi Lk_1} + \dfrac{\ln(r''/r')}{2\pi Lk_2} + \dfrac{\ln(r_o/r'')}{2\pi Lk_3}} \tag{2.17}$$

Equation (2.17) is of the form of Eq. (2.8). The total thermal resistance of the composite cylinder, given by the denominator of Eq. (2.17), is the sum of the thermal resistances of the individual layers. The overall temperature driving force is T_i - T_o. The electrical analogue of the problem is also shown in Fig. 2.4. The rate of heat conduction can be calculated if the cylinder dimensions, thermal conductivities of the layers and the driving force are known. Conversely, the temperature T' and T" can be obtained from the given or the calculated value of *Q*.

2.4.2. The Sphere

The temperature distribution in a hollow sphere or the rate of heat transfer through it can be determined by following the same procedure as used in the case of a cylinder.

Let us consider a hollow sphere of thermal conductivity *k* and inner and outer radii r_i and r_o respectively, with the corresponding surface temperatures T_i and T_o. Considering a thin spherical shell of inner radius r and thickness Δr (Fig. 2.5), we may write

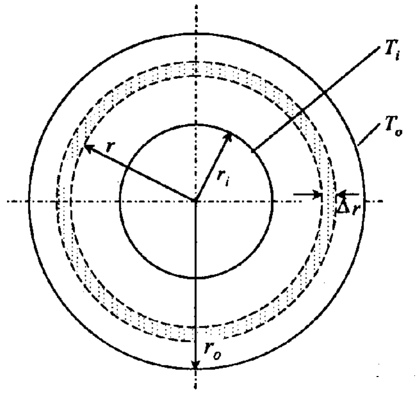


Fig. 2.5 Cross-section of the spherical shell

Rate of heat input at $r = 4\pi r^2 \, q_r|_r$

Rate of heat output at $r + \Delta r = 4\pi r^2 \, q_r|_{r+\Delta r}$

Because the rate of heat accumulation is zero at steady state, the heat balance equation is

$$4\pi r^2\, q_r|_r - 4\pi r^2\, q_r|_{r+\Delta r} = 0$$

or

$$\frac{4\pi r^2 q_r|_{r+\Delta r} - 4\pi r^2 q_r|_r}{\Delta r} = 0$$

Taking limit $\Delta r \rightarrow 0$, we get

$$-\frac{d}{dr}(r^2 q_r) = 0$$

or

$$r^2\, q_r = C_1 \qquad \textbf{(2.18)}$$

Putting $q_r = -k\dfrac{dT}{dr}$, and integrating, we get

$$T = \frac{C_1}{kr} + C_2 \qquad \textbf{(2.19)}$$

The boundary conditions are: $T = T_i$ at $r = r_i$, and $T = T_o$ at $r = r_o$. (2.20)

Using these boundary conditions in Eq. (2.19), the integration constants can be found as

$$C_1 = k\,(T_i - T_o)\left(\frac{r_i r_o}{r_o - r_i}\right) \quad \text{and} \quad C_2 = T_i - (T_i - T_o)\left(\frac{r_o}{r_o - r_i}\right) \qquad \textbf{(2.21)}$$

Therefore

$$T = T_i - (T_i - T_o)\left(\frac{r_o}{r_o - r_i}\right)\left(1 - \frac{r_i}{r}\right) \qquad \textbf{(2.22)}$$

Equation (2.22) gives the temperature distribution in the sphere. The corresponding rate of heat transfer is given by

$$Q = 4\pi r^2 q_r = 4\pi C_1 = \frac{T_i - T_o}{\dfrac{r_o - r_i}{4\pi k r_i r_o}} \qquad (2.23)$$

Comparing Eq. (2.23) with Eq. (2.8), we get

$$\text{Driving force} = T_i - T_o$$

$$\text{Thermal resistance} = \frac{r_o - r_i}{4\pi k r_i r_o}$$

If we rewrite Eq. (2.23) as

$$Q = k(4\pi\, r_i r_o)\,\frac{T_i - T_o}{r_o - r_i}$$

and compare it with Eq. (2.1), it appears that the rate of heat conduction through the spherical shell is the same as that through a plane wall of thickness ($r_o - r_i$) and area ($4\pi\, r_i\, r_o$). This is the area of the surface of a sphere of radius equal to the geometric mean radius of the given shell, i.e. $\sqrt{r_i r_o}$.

Similarly, if we consider a composite sphere having three layers of materials 1, 2 and 3, inner radii r_i, r' and r'', and thermal conductivities k_1, k_2 and k_3, respectively, and an outer radius r_o, (see Fig. 2.4), the rate of heat conduction through the composite sphere is given by

$$Q = \frac{T_i - T_o}{\dfrac{r' - r_i}{4\pi k_1 r' r_i} + \dfrac{r'' - r'}{4\pi k_2 r'' r'} + \dfrac{r_o - r''}{4\pi k_3 r_o r''}} \tag{2.24}$$

Equation (2.24) can be derived by following the same procedure as used in the case of a composite cylinder (it is left to the student as an exercise). If all other quantities are known, Q can be calculated, and if Q is known, the interface temperatures, T' and T" can be calculated.

2.5. APPLICATION OF CONDUCTION CALCULATION

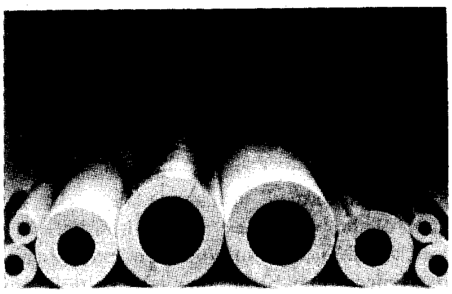
Heat conduction calculation is very important in a variety of situations. In some systems the rate of heat conduction is required to be as low as possible, whereas in some others a high rate of heat flow is desirable. For example, all furnaces are heavily insulated in multiple layers to minimize heat loss and to maintain a high temperature inside. The calculation of the rate of heat loss from a rectangular furnace can be done by using Eq. (2.7). Various process equipment, reactors, steam pipes and cryogenic systems need to be well-insulated against loss or gain of heat. Besides its most conspicuous function of reducing heat loss, thermal insulation on a hot pipe reduces the pipe stress. Electrical wires are always kept insulated [polyvinyl chloride (PVC) is a common electrical insulator used], and this layer of insulation also acts as a thermal insulator. If the steady state wire temperature is high, degradation of the insulation may occur. Thus, conduction calculation is important in electrical insulation design. Conduction calculation in nuclear fuel elements is also important to determine the size of the fuel element to be used so that the maximum temperature in it does not exceed the allowable limit. Determination of insulation thickness on steam pipes and various process equipment is a routine part of process design calculations. Important insulating materials used over different temperature ranges are listed in Table 2.2. The Petroleum Conservation Research Association (PCRA) gives an excellent account of the properties and applications of industrial thermal insulations.

Table 2.2 Common insulations used for process equipment and pipelines

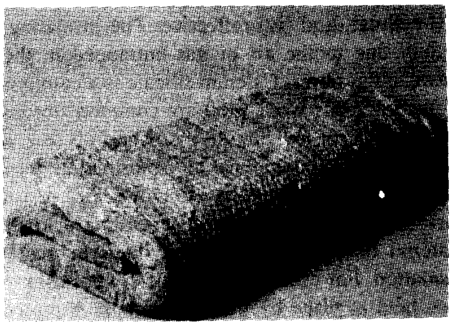
	Insulation material	*Temperature range* °C	*Thermal conductivity* (10^{-3} W/m °C)	(10^{-3} kcal/h m^2 °C)	*Density* (kg/m^3)	*Applications*
1.	Perlite	–200 to 40	1–2	0.85–1.7	60–140	Cryogenic services
2.	Polyurethane foam	–170 to 110	16–20	14–17	32	Tanks, vessels (cold/hot)
3.	Polyurethane foam	–180 to –150	16–20	14–17	25–50	Pipes and fittings
4.	Expanded polystyrene	–100 to 40	22–25	19–22	20–50	Chilled vessels
5.	Fibre-glass blanket	–160 to 230	24–86	21–75	10–50	Chillers, tanks, vessels
6.	Fibre-glass blanket for wrapping	–80 to 290	22–80	20–70	10–50	Pipes and fittings (hot/cold)
7.	Fibre-glass board	20 to 450	32–52	27–50	25–100	Boilers, tanks, heat exchangers
8.	Calcium silicate, board/block	230 to 1000	32–85	27–50	25–100	Boilers, breechings, chimney liners
9.	Mineral fibre block	Up to 1100	50–130	43–112	210	Boilers and tanks
10.	Fibre glass mats	60 to 360	30–55	26–47	10–50	Vessels, tanks
11.	Mineral fibre blankets	Up to 750	37–80	32–69	125	Vessels, pipes
12.	Mineral wool blankets	450 to 1000	50–130	43–112	175–290	Vessels, pipes

Common thermal insulators can be divided into four groups: (i) granular, (ii) fibrous, (iii) cellular, and (iv) reflective. The insulating property of materials of the first three groups is mainly due to the air or gas entrapped in the matrix (for granular and cellular materials) or between the fibres. Calcium silicate is a common granular insulating material. Mineral wools (e.g. rockwool, glass wool, etc.) are common fibrous insulations. A reflective insulation consists of a packed layer of low emissivity metal foils (e.g. aluminium) and is used for cryogenic applications.

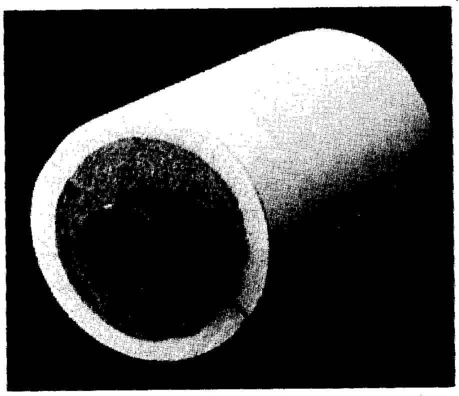
Selection and application of insulation depend upon a number of factors such as temperature of a body, its surface characteristics, fire resistance, etc. 'Pre-formed' insulations (Fig. 2.6) of various shapes may be used depending upon the shape of the body to be insulated. A bonding material like phenol-formaldehyde resin may be used for fabrication of rockwool pre-formed insulation. For example, two half-cylindrical pre-formed pieces may be put on a pipe section and then held at place by bands or straps. A protective cover, commonly an aluminium foil (called 'cladding'), is used to prevent damage or weathering of the insulation. Mineral wool 'mats' made by holding a layer of the material between two wire nets by stitching may be applied to a flat surface or a vessel wall. For cryogenic service, a coating of a suitable polymer formulation is applied on the outer surface of the insulation (e.g. polyurethane foam) to form a vapour barrier that prevents penetration of water vapour in the insulation.



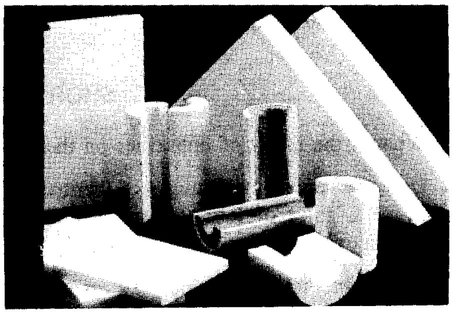
(a) A 'Rockloyd' pre-formed pipe section (made from bonded rockwool).



(b) A 'Rockloyd' mattress (made from light resin-bonded rock fibres, with wire mesh machine stitch on one side; used to insulate boilers, vessels, etc.).



(c) A bilayer pre-formed insulation (inner layer—rockwool, outer layer—polyurethane foam) on a cold surface.



(d) Pre-formed pieces of polyurethane foam (different shapes).

(e) Measurement of insulation surface temperature by a digital contact thermometer.

Fig. 2.6 Thermal insulations [Courtesy: Llyod Insulation (India), Calcutta].

PROBLEMS

2.1. An annealing chamber has a composite wall made of a 17 cm thick firebrick layer *(k* = 1.1 W/m °C) and a 13 cm thick ordinary brick layer *(k* = 0.70 W/m °C). The inside and outside surface temperatures of the walls are 400°C and 45°C, respectively. Calculate the heat loss from 25 m^2 of furnace wall. Also, determine the temperature between the ordinary brick and the firebrick layers.

2.2. The walls of a house in a cold region consist of three layers-an outer brickwork of 15 cm thickness and an inner wooden panel of 1.2 cm thickness. The intermediate layer is made of an insulating material 7 cm thick. The thermal conductivities of the brick and the wood used are 0.70 W/m °C and 0.18 W/m °C, respectively. The inside and outside temperatures of the composite wall are 21°C and -15°C, respectively. If the layer of insulation offers twice the thermal resistance of the brick wall, calculate (a) the rate of heat loss per unit area of the wall and (b) the thermal conductivity of the insulating material.

2.3. A 15 cm schedule 40 steam main carries saturated steam at 10.7 bar (gauge), and the temperature is 190°C. The inside and outside diameters of the pipe are 15.4 cm and 16.8 cm, respectively. The thermal conductivity of the pipe wall is 51 W/m °C. The pipe is insulated with a 10 cm thick fibre glass blanket *(k* = 0.072 W/m °C). If the outer surface temperature of the insulation is 41°C, calculate the rate of heat loss over a 10 m section of the pipe. Also, calculate the fraction of the total thermal resistance offered by the pipe wall. Is it justified to neglect the resistance of the metal wall in this type of problem?

2.4. A spherical vessel of 3 m inside diameter is made of AISI 316 stainless steel sheet of 9 mm thickness *(k* = 14 W/m °C). The inside temperature is - 80°C. The vessel is layered with a 10 cm thick polyurethane foam *(k* = 0.02 W/m °C) followed by a 15 cm outer layer of cork (k = 0.045 W/m °C). If the outside surface temperature is 30°C, calculate (a) the total thermal resistance of the insulated vessel wall, (b) the rate of heat flow to the vessel, (c) the temperature and heat flux at the interface between the polyurethane and the cork layers, and (d) the percentage error in calculation if the heat transfer resistance of the metal wall is neglected.

2.5. Consider a composite wall consisting of four different materials as shown in Figure 1. The following data are given: L_A = 0.1 m, L_C = 0.2 m, L_D = 0.12 m, H = 2 m, H_B = 1 m, k_A = 20 W/m °C, k_B = 10 W/m °C, k_c = 7 W/m °C, k_D = 25 W/m °C, T_1 = 120°C, and T_2 = 50°C.

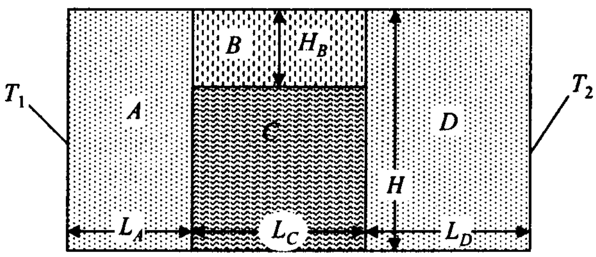


Figure 1 Composite wall (Problem 2.5)

Calculate the rate of heat flow through the assembly per unit breadth. Assume one dimensional heat flow only. Show the electric analogue of the problem.